Introduction to Computer Architecture and Systems

Chapter 0

Computer Architecture: Why and how?

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Motivation: Topics Forever

Programming with Constraints

Hardware + Software

For better Programmer

Hardware + Software

For better Computer Scientist

When and where for CSCI260?

Course Arrangements

- Class: Mondays, Wednesdays, and Fridays (MWF) 11:50 AM – 1:05 PM, at Brothers College (BC) 21, in person.
- Office Hours: Mondays and Wednesdays, 1:30 PM to 2:30 PM, at Hall of Sciences S311, in person or via Zoom. Feel free to make more arrangements.
- Holidays: Labor day (09/01), Fall break (10/06-07), and Thanksgiving (11/26-30).

What topics will we cover?

In this course, we will cover…

Foundation of Computer Architecture and Design:

  Key Concepts, Cognitive Elements…

Milestones of Computer Architecture and Design:

  Good, Bad, and Ugly…

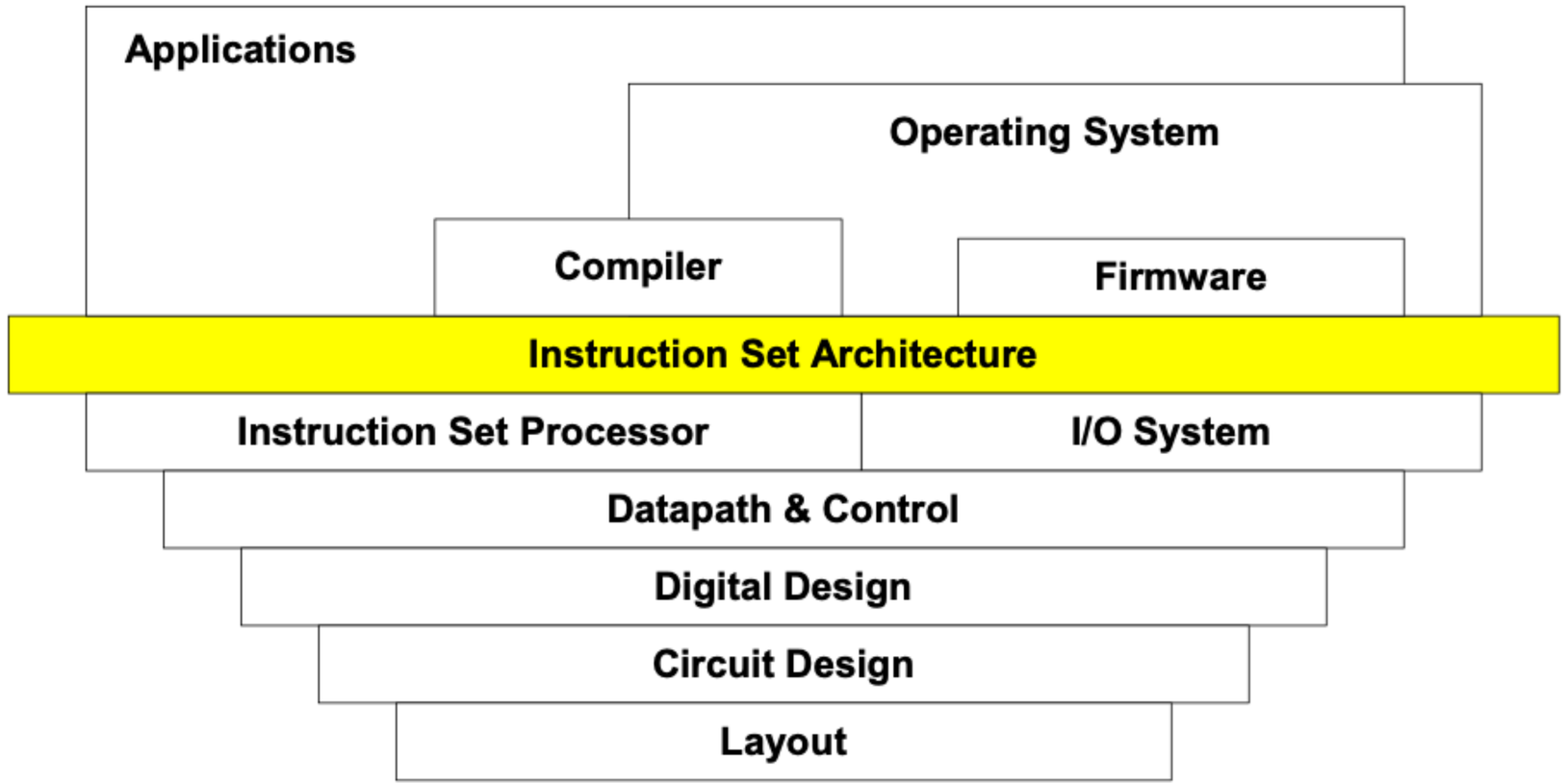
Relations between Computer Architectures and Systems:

  Great things on Hardware’s side, Software’s side, and both.

Programming and Researching considering Computer Architecture

Wait… where are we at?

Software



We are here

Hardware

Instruction Set Architecture

What is that? How it looks? Why do we need to learn it?

- The vocabulary of commands
- Defined by the Architecture (x86)
- Implemented by the Machine (Intel Core i7, 4 GHz)
- An Abstraction Layer: The Hardware/Software Interface
- Architecture has longevity over implementation
- Example:
- add r1 = r2 + r3 (assembly) -> 001 001 010 011 (binary)
  Opcode (verb) + Operands (nouns)

What will I get from CSCI260?

After CSCI 260, you will be able to…

- Appreciate theory vs. practice
  - Become a better implementer of algorithms
- Learn assembly/machine language programming
  - Essential for OS and Compiler Work
  - Essential for understanding processor design
- Understand modern Computer Organization
  - High-level languages -> execution on physical material
- Help you to revolutionize computing
  - Discuss the latest research
  - Contribute some of our own?

Introduction to ISA w Compilers

Java Program

```java
/* Hello World in Java */

public class HelloWorld {

    public static void main(String[] args) {
        System.out.println("Hello world!");
    }
}
```

C Program

```c
/*
 * Good programs have meaningful comments
 */
#include <stdio.h>


int main()
{
  printf("Hello World!\n");
  return 0;
}
```

```c
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int main()
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  printf("Hello World!\n");
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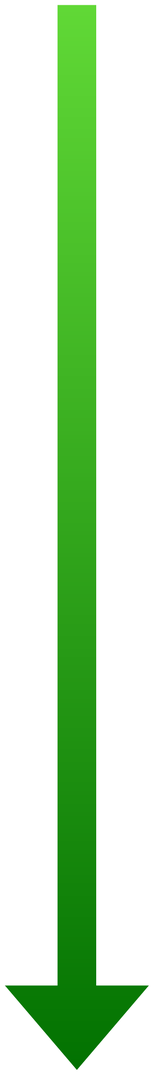
C Program → GNU C Compiler → x86 Assembly Language



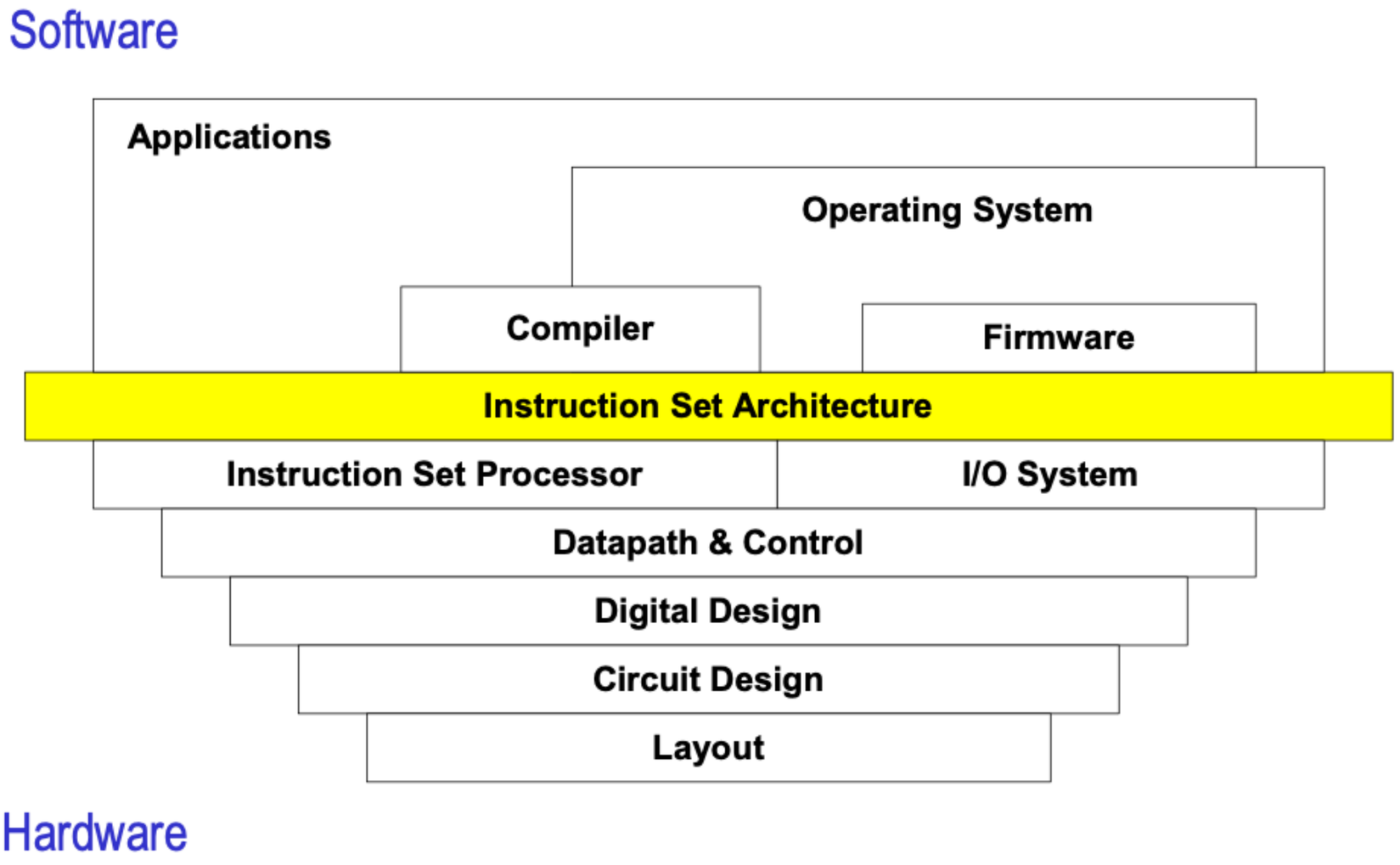
A compiler is a program that takes a program written in a source language and translates into a functionally equivalent target language

```
        .file   "hello.c"
        .section        .rodata
.LC0:
        .string "Hello World!\n"
        .text
.globl main
        .type   main,@function
main:
        pushl   %ebp
        movl    %esp, %ebp
        subl    $8, %esp
        andl    $-16, %esp
        movl    $0, %eax
        subl    %eax, %esp
        subl    $12, %esp
        pushl   $.LC0
        call    printf
        addl    $16, %esp
        movl    $0, %eax
        leave
        ret
.Lfe1:
        .size   main,.Lfe1-main
        .ident  "GCC: (GNU) 3.2.2 20030222 (Red Hat Linux 3.2.2-5)"
```

Things readable to users



Things readable to machines



Compiler make program runnable on your machine, by translating the code from the code you know, into the information your machine knows (ISA / Assembly).

Assignment & Assessment

In CSCI 260, you will have these assignments…

- 6 Assignments
- Midterm
- Assembly Project
- Final
- Quizzes

Quiz

Getting know about you…

Quiz 0: Background (use index cards)

All Quizzes are graded liberally.

- Front:
- 1. Full name and Email Address above the red line
- 2. Major/UG or G/Year (immediately below the red line)
- 3. Area (G: Research Area/UG: Interests)
- 4. Briefly describe any C/C++ experience.
- 5. In which programming languages are you fluent?
- 6. What is a bit?

Quiz 0: Background (use index cards)

All Quizzes are graded liberally.

- Back:
- 1. What is an instruction cache?
- 2. What is the difference between a sequential and a
- combinational circuit?
- 3. What is a MUX?
- 4. Using AND, OR, and NOT gates design an XOR gate.

Quiz 0: Background (use index cards)

- 1. What is an instruction cache?
  An **instruction cache (I-cache)** is a small, fast memory inside the CPU that stores recently used **instructions** (the code the CPU is executing), so the processor doesn't have to fetch them repeatedly from slower main memory (RAM).

- 2. What is the difference between a sequential and a combinational circuit?
  - Combinational = instant reaction, **no memory**.
  - Sequential = **has memory**, output depends on sequence of inputs over time.

- 3. What is a MUX?
  A MUX (Multiplexer) is a digital circuit that **selects one** input from many and forwards it to the output, based on control signals (also called select lines).

- 4. Using AND, OR, and NOT gates design an XOR gate.
  An XOR gate can be built using 2 NOTs, 2 ANDs, and 1 OR, with the Boolean expression:

$$A \oplus B = (A \cdot \overline{B}) + (\overline{A} \cdot B)$$